A Comparative Analysis of Deep Learning Models

for Fruit Ripeness Classification

CNN and Transformer Architectures

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Deep Learning · Computer Vision · Agriculture Technology

Presentation Overview

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Motivation & Problem Statement



Manual ripeness inspection is highly subjective, slow, and inconsistent



Correct ripeness detection is critical for harvest timing, logistics & food quality



Deep learning (CNNs & Transformers) can automate and standardize this process



Limited comparative studies exist between CNN and Transformer architectures for this task

Dataset & Preprocessing

Dataset

- Source: FreshCheck Fruit Ripeness DB
- Subject: Real-world tomato images
- 3 Classes: Unripe · Ripe · Overripe
- Captured via consumer cameras
- Varied lighting, angles, backgrounds
- Train / Test split: 80% / 20%

Preprocessing

- Resize all images to 224 × 224 px
- Normalize pixel values (x/255)
- Data augmentation: rotation, flipping
- Brightness adjustments
- Consistent format across all models
- Prevents overfitting on small dataset

Models Tested

9 deep learning architectures compared under identical conditions

CNN Architectures (6 Models)

Custom CNN ×3

Three custom-designed CNNs using conv, pooling & fully connected layers. Lightweight, fewer parameters.

DenseNet201

Each layer receives inputs from ALL prior layers — improves gradient flow & feature reuse.

AlexNet

Pioneer deep CNN. Demonstrated superiority of deep networks over traditional methods.

InceptionResNetV2

Combines Inception modules + residual connections. Captures multi-scale features efficiently.

Transformer Architectures (3 Models)

ViT-B/32	Vision Transformer — splits images into patches treated as sequences. Uses self-attention to model global dependencies.
DeiT-Small	Data-efficient Image Transformer. Uses distillation tokens to train effectively on smaller datasets.
Swin Transformer Tiny	Hierarchical transformer using shifted windows. Improves efficiency and scalability for vision tasks.

Results

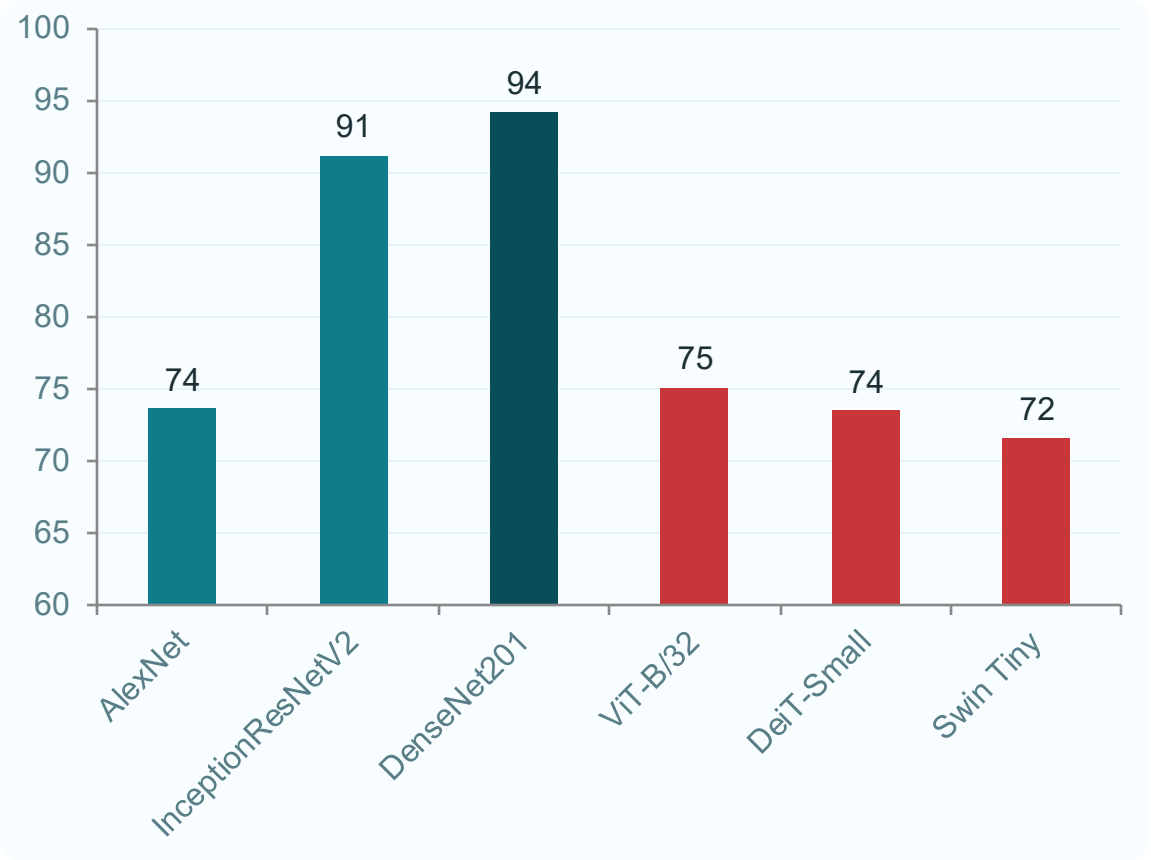
Accuracy · Precision · Recall · F1-Score

Overall Model Performance

Model	Accuracy	Precision	Recall	F1-Score
AlexNet	73.6%	78.6%	73.9%	68.7%
InceptionResNetV2	91.2%	91.2%	91.8%	91.6%
DenseNet201 ★	**94.2%**	**94.8%**	**94.0%**	**94.9%**
ViT-B/32	75.1%	76.7%	75.9%	71.3%
DeiT-Small	73.5%	71.1%	73.3%	69.2%
Swin Transformer Tiny	71.6%	61.2%	71.4%	65.3%

★ *Best performer | CNN models shown in rows 1–3 | Transformer models in rows 4–6*

CNN vs Transformer: Accuracy Comparison



Best CNN: DenseNet201
94.2% accuracy

Best Transformer: ViT-B/32
75.1% accuracy

Gap: ~19 percentage points

Key Findings

DenseNet201 Wins

Highest accuracy at 94.2% — dense layer connectivity enables efficient feature reuse & gradient flow

InceptionResNetV2 Runner-Up

91.2% accuracy — multi-scale inception modules + residual connections = strong feature extraction

Transformers Underperform

Best transformer (ViT-B/32) reached only 75.1% — ~19% gap behind the top CNN model

Why CNNs Win Here

Small dataset favors CNN's built-in spatial bias (locality). Transformers are data-hungry by design

Conclusion

- CNN architectures are significantly more effective than transformers for fruit ripeness classification on small datasets
- DenseNet201 achieved the highest accuracy (94.2%) — its dense connectivity is ideal for this visual task
- Transformer models require large datasets; their lack of spatial bias is a disadvantage here
- Custom lightweight CNNs performed competitively despite far fewer parameters
- Future work: larger datasets may allow transformers to close the performance gap